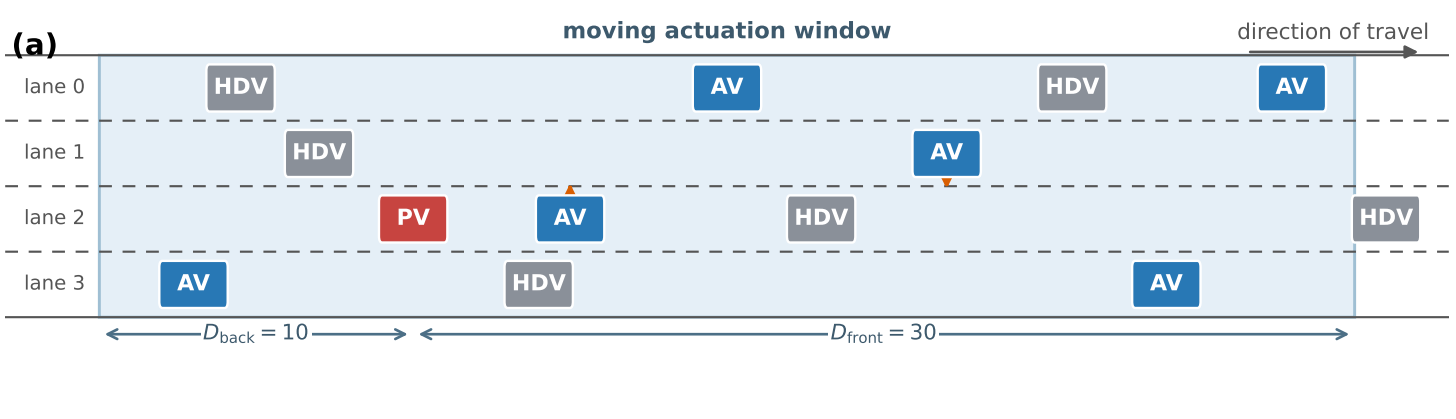
(a)
moving actuation window
direction of travel
lane 0
lane 1
lane 2
lane 3
HDV
AV
PV
$D_{back} = 10$
$D_{front} = 30$
(b)
PV-centred observation
48 node features
4 global features
action and node masks
Graph message passing
Nodewise MLP
Dilated grid CNN
shared actor
centralised critic
node-factorised PPO
masked lateral
requests
$-1, 0, +1$
Common reward, simulator, training distribution, and parameter budget (within 0.7%)